

Semantic Modeling for Supporting Planning Decision Making Towards Smart Cities

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ABSTRACT

The world is urbanizing rapidly. By 2050, 68% of the world population will reside in urban areas, posing unprecedented pressures on the livability, equity, and sustainability of city systems. While city smarterization holds great promise to release these pressures, the planning decision making towards smart cities is challenging because of the enormous breadth of smart city dimensions. As a result, decision makers are often lost in where and how to make the best investments to address the most pressing issues at hand. In recent years, many smart city initiatives and programs have been implemented, which offers an important source of smart city knowledge for decision makers to learn from. To this end, this paper proposes a smart city ontology, as a shared conceptualization of smart city knowledge, to facilitate the sharing and reuse of the knowledge for informed planning decision making. At the highest level of abstraction, the ontology semantically models three key elements of smart city knowledge, including visions, challenges, and solutions. This paper focuses on presenting the proposed ontology and its evaluation. The evaluation results showed the promise of the proposed ontology in supporting smart city planning decision making.

INTRODUCTION

Cities play a significant role in building a sustainable future (Mori and Christodoulou 2012). According to the United Nations, the percentage of the world population living in urban areas will increase from 55% in 2018 to 68% by 2050 (UN 2019). This unprecedented urbanization rate creates urgent challenges, such as aging society, traffic congestion, land loss, and environmental pollution, which reduce the quality of life in urban settlements and hamper the sustainable development of cities (Jnr et al. 2018). The concept of *smart city* is increasingly accepted in academia and policy making in a bid to tackle the challenges brought by rapid urbanization (Caragliu 2011). However, despite the promise of smart city, the planning decision making towards smart cities is challenging. Typically, smart city planning is concerned with deciding on the best use of social, economic, environmental, institutional, and technological resources to make

a city smart in all dimensions, in terms of living, economy, environment, governance, mobility, etc. (Junior et al. 2018). Because of the diversity of available resources and the breadth of smart city dimensions, planning decision makers are often lost in where and how to make the best smart city investments. What is missing is a precise picture of smart city visions, the associated challenges, and the potential solutions to these challenges to inform smart city planning decision making (USDOT 2015).

In recent years, many smart city initiatives and programs have been undertaken, such as the Smart City Challenge (USDOT 2016), the Smart City Initiative (City of Long Beach 2019), and the Smart Cities New York (Smart Cities NYC 2019). These initiatives and programs provide detailed planning and implementation reports to share the knowledge they learned, such as the smart city visions they aimed to achieve, the challenges they faced, and the solutions they adopted. They, thus, offer an important source of smart city knowledge for planning decision makers to learn from. However, smart city knowledge is often captured implicitly in textual sources using indirect language – rather than direct, explicit conceptualization – which hinders knowledge sharing and reuse. In addition, different textual sources often use different terminologies to refer to the same concept. For example, Boston and Birmingham use two different terms in their proposals to the Smart City Challenge (USDOT 2016) to refer to the same vision of reducing traffic accidents: “Zero death. Zero injuries” vs. “Improve traffic safety for all users”. There is, thus, a need for an explicit and shared conceptualization of smart city knowledge to better support smart city planning decision making.

To address this need, this paper proposes smart city ontology, as an explicit and shared conceptualization of smart city knowledge, to support smart city planning decision making. The ontology uses a concept hierarchy and cross-concept relationships to semantically model smart city knowledge in three main aspects: smart city vision knowledge (What are the visions in building a sustainable future?), smart city challenge knowledge (What are the challenges associated with a specific vision?), and smart city solution knowledge (What solutions can be used to address a specific challenge?). In the following, this paper focuses on presenting the proposed ontology and its evaluation.

BACKGROUND AND KNOWLEDGE GAPS

The concept of *smart city* was first proposed in the 1990s (Alawadhi et al. 2012). Ever since, many early studies have emphasized the role of technology in defining smart cities (Winkowska et al. 2019). As a result, a smart city was often referred to as a digital city, an information city, a ubiquitous city, or a sensing city (Albino 2015). In the contemporary perception, there is a strong consensus that a city cannot become a smart city solely through technological advancements, and a smart city should prioritize the needs and preferences of city inhabitants (Winkowska et al. 2020). This shift in the understanding of smart cities motivated studies to explore other dimensions beyond the technology dimension. For example, Giffinger and Gudrun (2010) defined six smart city dimensions, including smart economy, smart mobility, smart environment, smart people,

smart living, and smart governance. Lombardi et al. (2012) further detailed each of the six dimensions by defining its sub-dimensions, e.g., smart education and smart public event participation as sub-dimensions of smart people (Lobardi et al. 2012). In addition to the six dimensions, Albino et al. (2015) added the smart society dimension to smart city, which further captures the social inclusion of various urban residents and social capital in urban development.

Despite their importance, existing smart city studies are limited in supporting smart city planning decision making. Accordingly, two knowledge gaps were identified. First, there is a lack of studies that can provide deeper insights into smart city dimensions. Existing studies (e.g., Lobardi et al. 2012; Albino et al. 2015) mainly focus on high-level dimensions, without offering a deep conceptualization for these dimensions. As a result, decision makers can hardly establish tangible smart city visions to be achieved, if only relying on high-level dimensions (e.g., “smart mobility”) as opposed to detailed dimensions (e.g., “transit accessibility enhancement” and “travel length reduction”). Second, there is a lack of studies that provide a precise picture of the challenges towards achieving smart city visions and potential solutions for tackling these challenges. As a result, decision makers would encounter the situation that they only know the visions they aim to achieve, without a sense of the challenges ahead and the potential solutions to these challenges.

ONTOLOGY DEVELOPMENT METHODOLOGY

To address the aforementioned knowledge gaps, this paper proposes a smart city ontology to semantically model smart city knowledge in terms of visions, challenges, and solutions. The ontology development methodology by El-Gohary and El-Diraby (2010) was benchmarked to develop the smart city ontology. The following summarizes the main steps for developing the ontology:

- *Domain, purpose, intended users, and scope definition:* These fundamental attributes of the ontology were defined as follows: (1) domain: smart city domain; (2) purpose: modeling smart city knowledge to support planning decision making; (3) intended user: planning decision makers; (4) scope definition: covering concepts and relationships about smart city visions, challenges, and solutions.
- *Taxonomy construction:* The taxonomy was developed using two iterative steps: extracting main concepts from identified concept sources, and organizing the extracted concepts into a hierarchical hierarchy. The concepts were extracted from the smart city proposals submitted to the Smart City Challenge (USDOT 2016). The Challenge is organized by the U.S. Department of Transportation to invite medium-sized cities to put forward implementable solutions for addressing the most pressing challenges they are facing (USDOT 2016). A total of 78 cities submitted smart city proposals to the Challenge. Each proposal presents the existing characteristics of the city, the visions to be pursued, the challenges to achieving the visions, and the potential solutions in response to the

anticipated challenges. The extracted concepts were then structured into a taxonomic hierarchy using a combination of top-down and bottom-up approaches.

- *Relation modeling*: Three types of relationships were modeled in the ontology, including “is-a” relationships, “is-part-of” relationships, and cross-concept relationships.
- *Ontology coding*: The ontology was coded in Web Ontology Language using Protégé 5.5.0 (which is an open-source ontology editor).
- *Ontology evaluation*: The evaluation of the ontology is discussed in the “Ontology Evaluation” section.

PROPOSED SMART CITY ONTOLOGY

Main ontology model. Figure 1 shows the main model, which represents the most abstract concepts of the ontology. At the highest level of the abstraction, a “city” *has* a “smart city vision”, which *is associated with* a “smart city challenge” that *could be addressed by* a “smart city solution”. In addition, A “city” *is associated with* a “smart city dimension”, which *defines* a “smart city vision”, a “smart city challenge”, and a “smart city solution”. For example, “quality of life” as a smart city dimension concept defines a vision concept “quality of life enhancement”, a challenge concept “qualify of life enhancement challenge”, and a solution concept “quality of life enhancement solution”. A total of nine dimensions were defined in the main model.

- *Quality of life*: Qualify of life refers to the degree of satisfaction of an individual or a group with health, comfort, and life event participation.
- *Mobility*: Mobility refers to the ability to move from one point to another quickly and easily.
- *Infrastructure*: Infrastructure refers to the basic physical structures and facilities required to maintain the normal operation of society.
- *Sustainability*: Sustainability refers to meet the needs of the present, without compromising the needs of future generations.
- *Economy*: Economy refers to the wealth and resources of an area, particularly in terms of the production and consumption of commodities and services.
- *Governance*: Governance refers to all of the processes of governing. It relates to the processes of interaction and decision making among everything involved in the event.
- *Social cohesion*: Social cohesion refers to the connectedness and solidarity among groups in society.
- *Land use*: Land use refers to the human use (e.g., management and modification) of the natural environment or wilderness for developing built environments.
- *Technology*: Technology refers to scientific knowledge applications for practical purposes, especially in the engineering area.

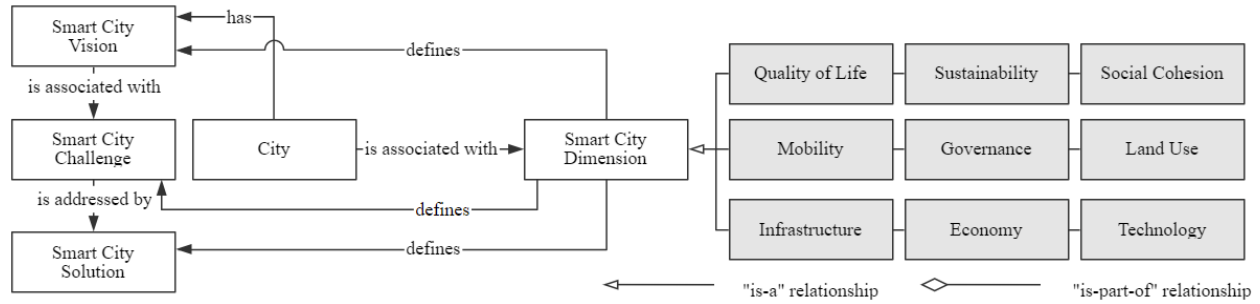


Figure 1. Main smart city ontology model.

Smart city vision hierarchy. Figure 2 shows a partial view of the smart city vision hierarchy. At the highest level of abstraction, a “smart city vision” could be a “quality of life enhancement”, a “mobility performance enhancement”, an “infrastructure optimization”, a “sustainability promotion”, an “economic vitality enhancement”, a “better governance”, a “social cohesion enhancement”, an “efficient land use”, or a “smart technology”. This hierarchy models different types of detailed subconcepts to provide an in-depth conceptualization of smart city visions. For example, a “qualify of life enhancement” could be a “safety enhancement”, a “homeless eradication”, a “health enhancement”, or a “knowledge transfer enhancement”. A “sustainability promotion” could be an “energy conservation”, a “renewable energy source”, or an “environment impact reduction”.

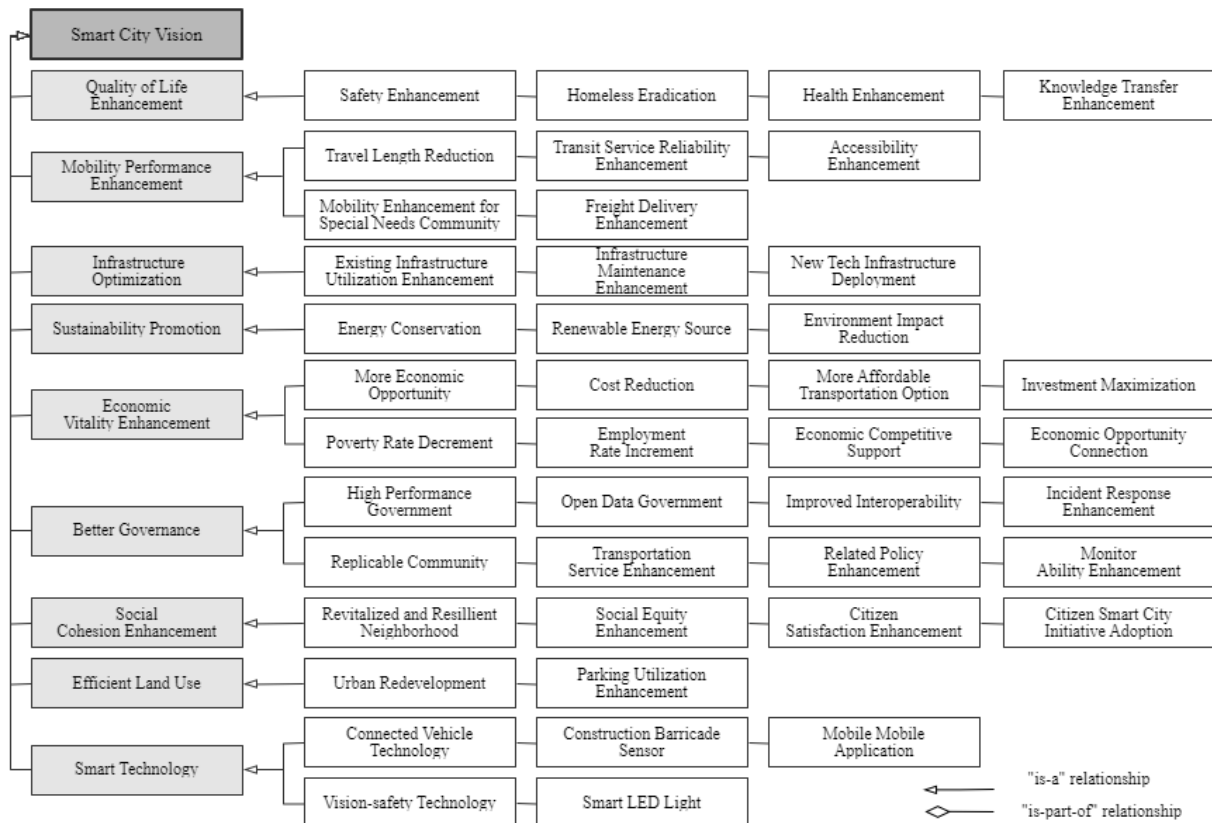


Figure 2. Partial view of the smart city vision hierarchy.

Smart city challenge hierarchy. Figure 3 shows a partial view of the smart city challenge hierarchy. At the highest level of abstraction, a “smart city challenge” could be a “quality of life challenge”, a “mobility challenge”, an “infrastructure challenge”, a “climate change and sustainability challenge”, an “economic challenge”, a “land use challenge”, or a “lack of data sharing challenge”. This hierarchy models different types of detailed subconcepts to provide an in-depth conceptualization of smart city challenges. For example, a “quality of life challenge” could be a “population challenge”, a “safety challenge”, an “education challenge”, a “health challenge”, and a “social inequity challenge”. A “climate change and sustainability challenge” could be an “air quality challenge”, or an “environment protection challenge”.

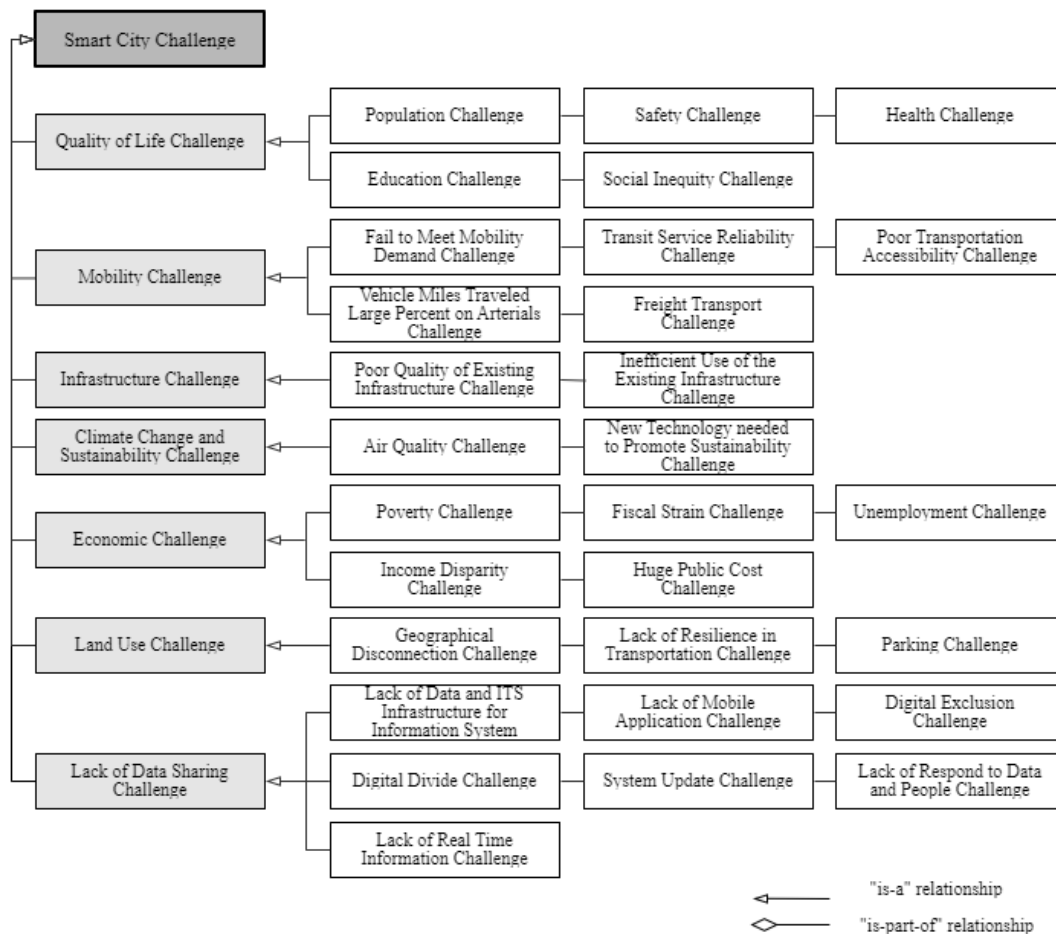


Figure 3. Partial view of the smart city challenge hierarchy.

Smart city solution hierarchy. Figure 4 presents a partial view of the smart city solution hierarchy. At the highest level of abstraction of this hierarchy, a “smart city solution” could be a “quality of life enhancement solution”, a “mobility performance enhancement solution”, a “sustainability promotion solution”, a “smart land use solution”, a “transportation infrastructure enhancement solution”, or a “smart technology solution”. The hierarchy represents different levels of the smart city solutions. For example, a “quality of life enhancement solution” could be a “safety

enhancement solution” and a “knowledge transfer enhancement solution”. An “infrastructure enhancement solution” could be a “road and pedestrian enhancement solution”, a “maintenance decision support system”, or an “intelligent sensor-based infrastructure”.

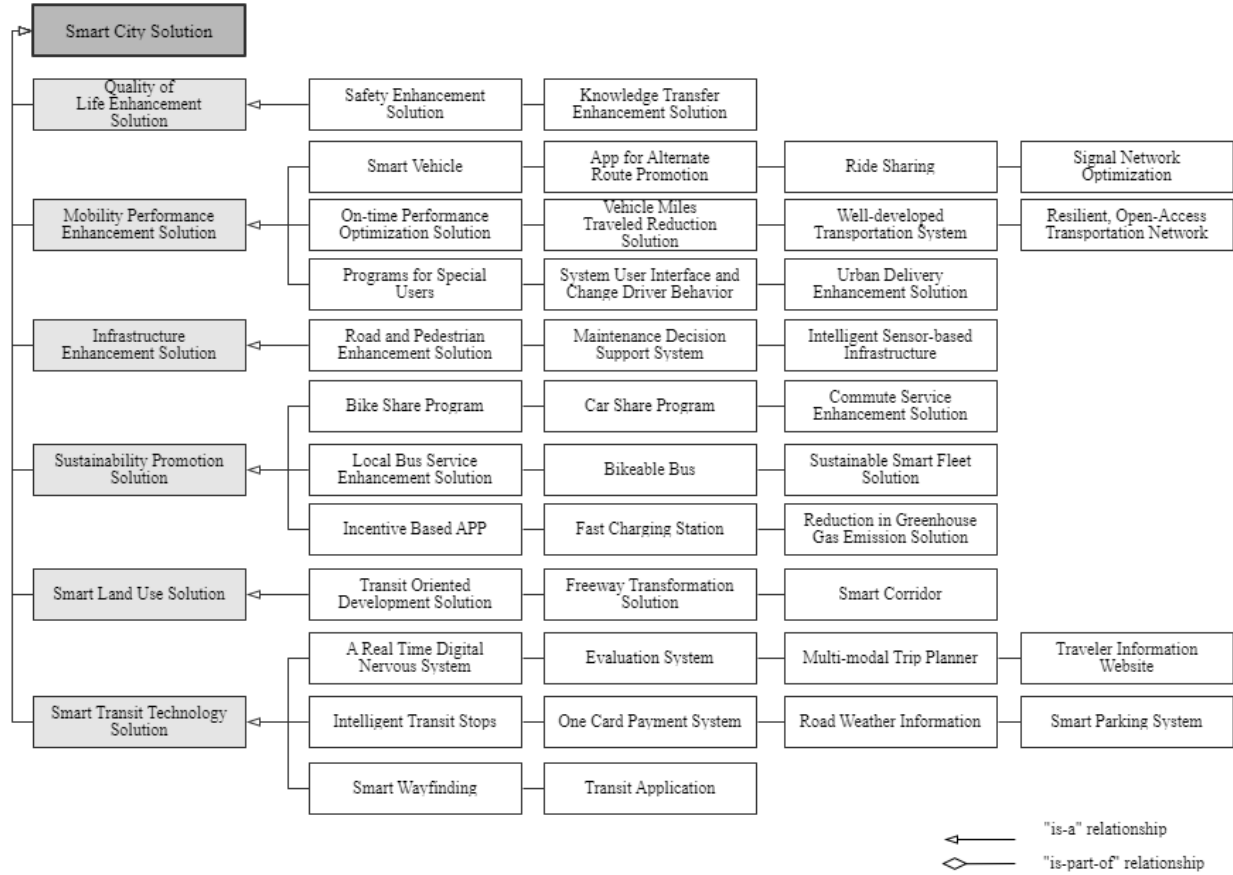


Figure 4. Partial view of the smart city proposed solution hierarchy.

Cross-concept relationships. In the ontology, cross-concept relationships were modeled to capture the semantic relationships between smart city vision, challenge, and solution concepts. Cross-relationships are also very important to inform planning decision making, because they allow for linking a vision to a specific set of challenges and linking a particular challenge to a set of potential solutions. Figure 5 shows a partial view of the cross-concept relationships. For example, the vision “existing infrastructure utilization enhancement” *is associated with* “poor quality of existing infrastructure challenge” and “inefficient use of existing infrastructure challenge”, which *could be addressed by* “road and pedestrian enhancement solution” and “maintenance support decision system”, respectively. The vision “urban redevelopment” *is associated with* challenges such as “geographical disconnection challenge” and “lack of resilience in transportation challenge”, which *could be addressed by* “transit-oriented development solution” and “freeway transformation solution”, respectively.

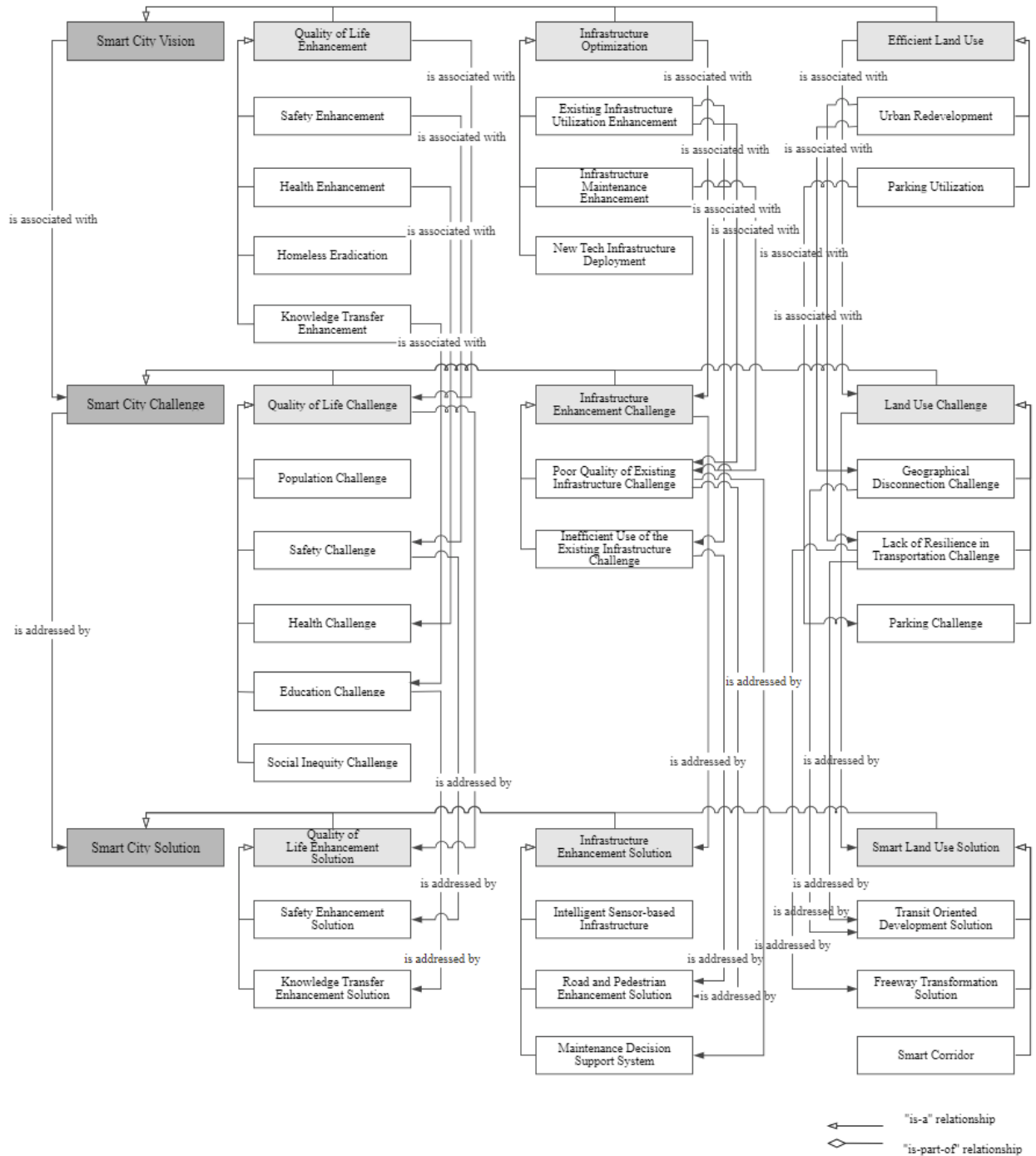


Figure 5. Partial view of the cross-concept relationships.

ONTOLOGY EVALUATION

Ontology evaluation typically includes verification and validation. In this paper, two verification methods were used to evaluate the smart city ontology: (1) answering competency questions, which aimed to verify if the ontology is able to answer all the required questions; and (2) automated

consistency and redundancy checking, which aimed to verify if the ontology is free of consistency and redundancy errors. A set of competency questions were developed by the authors. An example question is “What are the challenges associated with the vision of health enhancement?” The ontology was able to answer all the questions. The Pellet 1.5.2 Reasoner, which is a built-in reasoner in Protégé 5.5.0, was used to conduct the checking. The checking results showed that the ontology does not have inconsistencies and redundancies. The validation of the ontology is discussed in the “Conclusion and Future Work” section.

CONCLUSION AND FUTURE WORK

This paper proposed an ontology to semantically model smart city knowledge for supporting smart city planning decision making. The ontology captures the knowledge of smart city visions, challenges, and solutions, with in-depth conceptualization and classification as well as detailed semantic relationships. The proposed ontology could serve as an explicit shared specification to support the sharing and reuse of smart city knowledge for improved planning decision making. In this preliminary study, the ontology was verified through answering competency questions and automated consistency and redundancy checking. The authors’ ongoing work is focusing on (1) enriching the ontology by also modeling smart city concepts from other concept sources to improve its coverage; and (2) conducting ontology validation using expert interviews and application-oriented evaluation. In their future work, the authors plan to devote their efforts in two directions: (1) conducting empirical studies to evaluate the satisfaction levels of city inhabitants with the current smartness level of the city and the planning decisions; and (2) developing data-driven methods to quantify and predict the effectiveness of planning decisions in smaterizing cities. These efforts would allow for engaging inhabitants into the loop of city planning decision making to develop more inclusive smart cities.

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